

INTOXICOM Workshop Report: Making toxicology tools more accessible and interoperable

Meike Bünger ¹, Ola Spjuth ^{2,7}, Nikita Churikov ³,
Jonne Rietdijk ^{2,7}, Jente Houweling ^{4, 5}, Wolmar Nyberg
Åkerström ^{6, 7}, Ogunleye Adeolu ⁸, Penny Nymark ⁹, Petru
Niga ¹⁰, Cleo Tebbi ¹¹, and Egon Willighagen ⁵

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1 Stichting Health-RI, Utrecht, The Netherlands 
2 Dept of Pharmaceutical Biosciences, Uppsala University, Uppsala, Sweden 
3 Dept of Immunology, Genetics and Pathology, Uppsala University, Uppsala, Sweden 
4 RIVM, Bilthoven, The Netherlands 
5 Dept of Translational Genomics, Maastricht University, Maastricht, The Netherlands 
6 National Bioinformatics Infrastructure Sweden, Sweden 
7 Science for Life Laboratory, Sweden 
8 Dept of Organismal Biology, Uppsala University, Uppsala, Sweden 
9 Institute of Environmental Health, Karolinska Institutet, Stockholm, Sweden 
10 Chemical Processes and Pharmaceutical Development, RISE Research Institute of Sweden., Sweden 
11 New approaches and models for human health and environmental safety unit, Ineris, Verneuil-en-Halatte, France

Introduction

As part of the INTOXICOM Implementation Study for the ELIXIR Toxicology Community a series of workshops is organized (Martens et al., 2024). Here, we here report on the 3rd workshop, titled “Making toxicology tools more accessible and interoperable” which was held from 26 to 27 March 2025 at the SciLifeLab at Uppsala University in Sweden.

The workshop welcomed 29 participants from Sweden, The Netherlands, Cyprus, Switzerland, Italy, Greece, France, and Norway. This 3rd INTOXICOM workshop covered various aspects of making computational tools available and how they are used. Several projects, such as NanoSolveIT (Afantitis et al., 2020), ONTOX (Vinken et al., 2021), and VHP4Safety (Kienhuis et al., 2024), already make computational toxicology services available, but the field of integrating computational toxicology dates back much longer, such as Bioclipse developed at Uppsala University (Willighagen et al., 2011), making it a perfect location to have held this workshop.

Presentations

Organizers and invited speakers held several presentation to provide information to support the workshop, scheduled as part of sessions hosted by Ola Spjuth and Egon Willighagen:

Speaker	Talk Title
Egon Willighagen	ELIXIR Introduction
Jonne Rietdijk	User perspective: Morphological Profiling for Chemical Toxicity Assessment
Ivo Djidrovski	Large Language Model tools and AI-Agents in toxicology research
Antreas Afantitis	NovaMechanics platforms: NanoSolveIT, Enalos Cloud, Pharos DMS
Ziye Zheng	Toxicity prediction with regulatory relevance: predicting the CLP H-statement
Egon Willighagen	Toxicity prediction with regulatory relevance: predicting the CLP H-statement (Willighagen, 2025)

Speaker	Talk Title
Nikita Churikov	Sharing and serving ML models in toxicology research with SciLifeLab Serve
Jente Houweling	VHP4Safety project for data/tool integration

Results

Day 1

After a *Welcome at SciLifeLab* by Ola Spjuth and an overview of ELIXIR Europe, the INTOXICOM workshop, and the schedule of the workshop, the first session was about *What tools for toxicology exist?*. The focus of this session was aimed at relevant services indexed by the ELIXIR Toxicology Community on bio.tools for computational tools and FAIRsharing for databases:

- [bio.tools records annotated with toxicology](#)
- [FAIRsharing's toxicology collection](#)

The participants that several services were missing in the collection of 103 tools found in bio.tools, including [VHP4Safety services](#), [TXG-MAPr](#), [BMDEExpress](#), the [Enalos Cloud Platform](#), OPERA and VEGA, [EFSA tools](#), and <https://www.parc-models.eu/tools/>.

The FAIRsharing collection was already discussed at the second INTOXICOM workshop (Bonatto Minella et al., 2026) and further curated by Bonatto Minella *et al.* (Bonatto Minella et al., 2025). During the workshop several databases were found to be missing, including ToxVal, ToxRef, ToxCast, [AOP-Wiki](#), [ECHA CHEM](#), [Nanosafety Data Interface](#), [GenRA](#), [chemPharos](#), [RepDose Database Fraunhofer ITEM](#), and [Le Portail Substances Chimiques \(PSC\)](#).

In the afternoon, the second session consisted of three user stories on data and tools in the session *How to make your data integrate with tools?*, chaired by Ola. The first story was given by Jonne Rietdijk on morphological profiling for chemical toxicity, proposed for toxicology testing in 2010 (Krewski et al., 2010). Rietdijk walked the participants through various approaches, such as cell painting. Data was shared on Figshare (Rietdijk et al., 2022), comparing effects of exposure to cetyltrimethylammonium bromide, bisphenol A, and bibutyltin dilaurate in several human cell lines. Images can be shared with OMERO (Allan et al., 2012), which has advantages over the more general Figshare.

The story provided by Ivo Djidrovski describes the rapid progress of large language models (LLMs). His work focuses on using LLM agents capabilities, which provide access to data and knowledge bases. He demos the use of LLMs in a chemical hazard assistant, based on the OECD QSAR Toolbox (Djidrovski et al., 2026).

After a break, Antreas Afantitis describes the NovaMechanics platforms NanoSolveIT, Enalos Cloud, Pharos DMS, based on 20 years of experience in the field. [Enalos cloud](#) enables non-expert users to leverage ML. Model outputs are accompanied by a workflow of what the user has performed. The [Nanopharos](#) database integrates experimental data with physics-based models and machine learning.

The last user story was provided by Ziyi Zheng for the IVL Swedish Environmental Institute. It discussed [SafeChem](#), the regulatory relevance, lack of negative controls, data mining from the ECHA REACH dossiers, and computational toxicology approaches, including molecular descriptors, fingerprints, and descriptor-free approaches, and comparing traditional machine learning and deep learning.

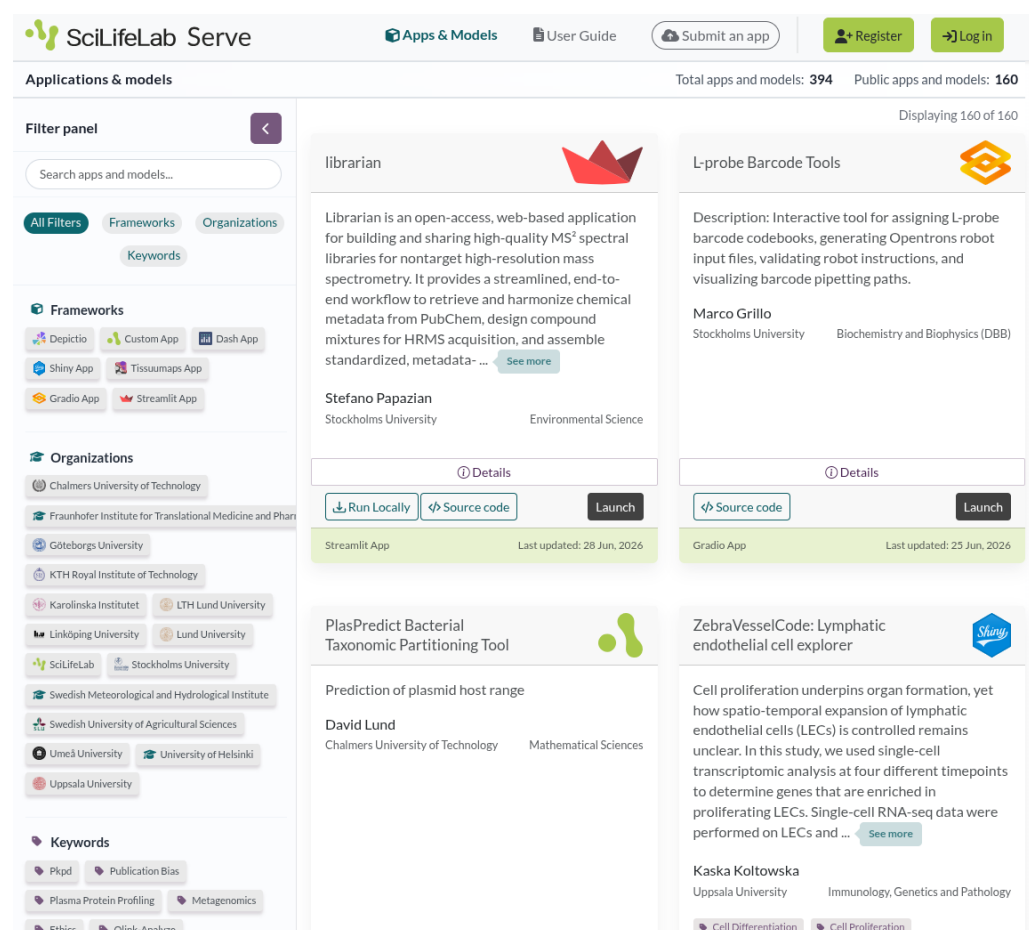
During the discussion at the end of this session, the workshop participants identified several missing things (some gaps are filled by the wider community in the year following the work-

shop): model context protocols (MCPs) for toxicology, Open API definitions, clarity and/in documentation, also for knowledge graphs, and better interoperability standards, for example, with FAIR Implementation Profiles.

Day 2

The second day started with the session *How can solutions complement your data? Using tools in tox studies* with three presentations and an open discussion. The first presentation was given by Egon Willighagen on how knowledge bases give context to experimental data (Willighagen, 2025). For toxicology, these may include WikiPathways (Agrawal et al., 2023), and AOP-Wiki (Martens et al., 2022), e.g. via the AOP-Wiki RDF. The importance of knowledge bases is that they curate the collected scientific knowledge. Bio2RDF gets a shout out for making knowledge bases interoperable (Belleau et al., 2008).

The second talk was given by Nikita Churikov of SciLifeLab on *Sharing and serving ML models in toxicology research with SciLifeLab Serve*. [Serve](#) is an infrastructure for bioinformaticians to publish open source codes (see Figure 1). The platform uses Docker to make tools available, such as DICTrank Predictor (Seal et al., 2024). The talk further discussed exchange of metadata and minimal reporting standards for tools and the [Five recommendations for FAIR software](#) and the [Self-assessment for FAIR research software](#) come up.



The screenshot displays the SciLifeLab Serve website interface. At the top, there's a navigation bar with the SciLifeLab Serve logo, links for 'Apps & Models', 'User Guide', 'Submit an app', 'Register', and 'Log In'. Below the navigation bar, a header indicates 'Applications & models' with a count of 'Total apps and models: 394' and 'Public apps and models: 160'. A 'Filter panel' on the left allows users to search and filter by 'Frameworks' (listing various app types like Depictio, Custom App, Dash App, Shiny App, TissueMaps App, Gradio App, and Streamlit App) and 'Organizations' (listing various institutions like Chalmers University of Technology, Fraunhofer Institute, Göteborgs University, KTH Royal Institute of Technology, Karolinska Institutet, LTH Lund University, Linköping University, Lund University, SciLifeLab, Stockholm University, Swedish Meteorological and Hydrological Institute, Swedish University of Agricultural Sciences, Umeå University, University of Helsinki, and Uppsala University). It also includes a 'Keywords' section with tags like Pkpd, Publication Bias, Plasma Protein Profiling, Metagenomics, Ethics, and Olink-Analyze. The main content area displays a grid of application cards. Each card includes the app name, a description, the author's name and affiliation, and buttons for 'Details', 'Run Locally', 'Source code', and 'Launch'. The cards shown are 'librarian' (Stefano Papazian, Environmental Science), 'L-probe Barcode Tools' (Marco Grillo, Biochemistry and Biophysics (DBB)), 'PlasPredict Bacterial Taxonomic Partitioning Tool' (David Lund, Mathematical Sciences), and 'ZebraVesselCode: Lymphatic endothelial cell explorer' (Kaska Koltowska, Immunology, Genetics and Pathology).

Figure 1: Screenshot of the SciLifeLab Serve website.

The last talk in this session was given by Jente Houweling working at the Dutch [National Institute for Public Health and the Environment](#). She introduces the participants to the

VHP4Safety project and her research on the [ToxTempAssistant](#) (Houweling et al., 2026) in particular. This tool uses large language models to support filling out ToxTemp templates, aimed at providing sufficient details on experimental toxicology data (Krebs et al., 2019).

Discussion

All sessions has many lively discussions with a lot more pointers and actions. Of particularly interest is the interest in using the interoperability that toxicologists have worked on over the years to enable LLMs to interface with knowledge bases and tools. This interest is shared by many, and ELIXIR Europe now has an [AI Ecosystem Focus Group](#), which includes two task forces, one on agentic AI and the other on generative AI.

Participants expressed concern about how and to which extent in silico tools should and can be validated, including tools that implement LLM models. Traceability of the training data, in particular that used for complex models should be ensured so that this data is not used, at later stages of model interoperability, to evaluate model or tool performance

Another important point is that finding tools and knowledge bases is a continuous activity, and the Toxicology Community will continue have to be active in ensuring they are easily found (as part of FAIR). During the workshop several tools and databases were identified which were not found in bio.tools of FAIRsharing at the time, and they need to be added. Platforms like VHP4Safety, NanoSolveIT, and SciLifeLab Serve should make metadata interoperable, e.g. with BioSchemas.

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Author contributions

Egon Willighagen: Writing – review & editing